

## Hyper Text Markup Language (HTML):

- > HTML is the standard markup language for creating web pages.
- > HTML describes the structure of a web page.

<html> represents the root of an HTML document

## Extensible Markup Language (XML)

- > XML stands for Extensible Markup Language.
- > XML was designed to store & transport data.
- > XML was designed to be both human- and machine-readable.
- > XML is a markup language much like HTML.
- > XML was designed to be self-descriptive.

## Domain Names

- > A domain name is a website's address on the Internet.
- > Domain names are used in URLs to identify to which server belong a specific webpage.
- > Domain name consist of a hierarchical sequence of names (labels) separated by periods (dots) and ending with an extension.

## URL

- > Uniform Resource Locator (URL) is a text string that specifies where a resource such as a web page, image or video can be found on the Internet.



## Website

- > Website is a group of web pages, containing text, images & all types of multi-media files.

## Web browser

- > A web browser or simply "browser", is an application used to access & view websites. Common web browsers include Microsoft Internet Explorer, Google Chrome, Mozilla Firefox and Apple Safari.

## Web server

- > A web server is a computer that stores web server software & a website's components like eg. HTML documents, images, CSS style sheets, Java Script files.

## Web hosting

- > Web hosting is an online service that enables you to publish your website via web application on the internet. When you sign up for a hosting service, you basically rent some space on a server on which you can store all the files & data necessary for your website to work properly.





> H17P5 (Secure Hyper Text Transfer Protocol)

It is an extension of H17P protocol for transmitting hypertext documents, such as HTML, securely over a network.

• It encrypts data to be sent using TLS (Transport Layer Security) SSL (Secure Socket Layer).

> Telnet (Terminal Network)

It is an application protocol that allows a user to communicate with a remote device.

> SMTP (Simple Mail Transfer Protocol)

It is used to send mail from mail client to mail server over internet.

> POP3 (Post Office Protocol)

It provides mechanism for retrieving emails from a remote server for a mail recipient.

• POP3 Downloads the email from a server to a single computer, then deletes the email from the server.

> VoIP (Voice over IP)

It is also known as Internet Telephony or Internet calling.

It allows to make voice calls using a broadband Internet connection instead of a regular (or analog) phone line.

ii) Introduction to web services

WWW (World Wide Web)

> HTTP (Hypertext Transfer Protocol)

It is the internet standard for transmission of IP packets over the serial lines. It is currently best solution for end-to-end connections.

iii) Introduction to web services  
WWW (World Wide Web)

> The web is an information system where documents and other web resources are identified by Uniform Resource locators, which maybe interlinked by hypertexts, are accessible over the Internet.

> Tim Berners-Lee proposed the architecture WWW in 1989.



## Internet

It is the global network of interconnected servers that may link and follow same set of rules, and connect together for sharing info and establishing communication.

## 1) Types of Network

### PAN

- > It stands for Personal Area Network
- > It is a computer network formed around a person
- > It includes mobile devices, tablet & laptop

### LAN

- > It is a group of computer and peripheral devices which are connected in a limited area such as room, building & campus.
- > Higher data speed
- > It is a private network

### MAN

- > A Metropolitan Area Network or MAN is consisting of a computer network that spans across a city
- > It mostly covers towns & cities in a maximum 50 km range.
- > The dual bus in MAN network provides support to transmit data in both directions concurrently.

### WAN

- > It connects device across globe
- > It uses public network
- > Internet
- > BSNL

## 2) Network Protocols

Protocol - It is a set of rules or standard that governs communication

Types of Protocols (Broadly can be kept in two series of Protocols viz. TCP/IP and OSI)

### > FTP (File Transmission Protocol)

It is used for the transfer of computer files among hosts over TCP/IP (Internet)

### > TCP/IP - Transmission Control Protocol / Internet Protocol

It is a protocol suite consist of two protocols TCP & IP  
TCP ensures reliable transmission or delivery of packets on the network.

### > HTTP (Hypertext Transfer Protocol)

It is an application-layer protocol for transmitting hypertext documents such as HTML



## C.S. ASSIGNMENT

### D) Evolution of Networking

#### Introduction to Computer Networks

Computer Network is a collection of autonomous computers interconnected by a single technology. Two computers are said to be interconnected if they are able to exchange info.

#### > ARPANET

(Advanced Research Project Agency Network)

In the year 1967, the very first network came into existence, namely - ARPANET.

It was designed to survive any nuclear threat.

It was the first system to implement the TCP/IP protocol suite and was based on Client-Server architecture.

#### > NSFNet

National Science Foundation Network was started in 1980 with a view to enhance Academic and Scientific Research. It connected its server with the ARPANET in the year 1986.

In the year 1990, the NSFNet, ARPANET and other smaller networks clubbed together to form the INTERNET (Interconnected Networks) and hence the foundation of modern INTERNET was laid down.